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[22CE31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

ENGINEERING GEOLOGY AND ITS APPLICATIONS IN CIVIL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. ANSWER ALL QUESTIONS

PART-A **5 X 2=10**

1.	a.	Delineate the presumptive basis of internal structure of earth.	L2	CO1	02
	b.	Differentiate between conglomerate and breccia.	L3	CO2	02
	c.	What are the distinguishing properties of Chalcopyrite?	L2	CO3	02
	d.	How is a dam is different from reservoir?	L2	CO4	02
	e.	What are the pre-requisites for a rock formation to become an aquifer?	L2	CO5	02
		PART-B			**6 X 5=30**
2.	a.	Describe the seismic zones of India.	L2	CO1	05
		OR			
	b.	How the earthquake intensity is measured?	L3	CO1	05
3.	a.	Give the classification of rocks and explain briefly	L2	CO2	05
		OR			
	b.	Discuss the textural classification of Igneous Rocks.	L2	CO2	05
4.	a.	Explain the physical properties of any two carbonate group minerals. Add a note on their importance.	L2	CO3	05
		OR			
	b.	Write short notes on feldspar group minerals and its applications.	L2	CO3	05
5.	a.	Explain the types and mechanism of folding. How folds are influencing the civil engineering structures?	L2	CO4	05
		OR			
	b.	How are the geological investigations for a good dam site are conducted?	L2	CO4	05
6.	a.	Illustrate hydrological cycle with a neat diagram.	L2	CO5	05
		OR			
	b.	Rainwater harvesting is an option for raising Ground water table, explain	L2	CO5	05

7.	a.	What are the basics for selection of rocks as construction materials?	L4	CO6	05
		OR			
	b.	What are the problems of faults in civil engineering works?	L4	CO6	05
		PART-C		**6 X 10=60**	
8.	a.	Describe internal structure and composition of the earth with a neat sketch.	L2	CO1	10
		OR			
	b.	What are the engineering problems related to earthquakes? How to overcome from such problems?	L4	CO1	10
9.	a.	Explain the classification of Sedimentary Rocks.	L2	CO2	10
		OR			
	b.	Define Metamorphism. Describe the kinds of metamorphism.	L2	CO2	10
10.	a.	Explain any two important physical properties of minerals and their classification for mineral identification.	L2	CO3	10
		OR			
	b.	Explain the verities of quartz and its uses.	L2	CO3	10
11.	a.	What are the different types of unconformities? Add a note on criteria for recognition and Engineering considerations of unconformity.	L3	CO4	10
		OR			
	b.	Discuss the problems associated with dam sites.	L2	CO4	10
12.	a.	Explain the geophysical investigation for groundwater exploration. Add a note on limitations of geophysical exploration.	L2	CO5	10
		OR			
	b.	Explain the different methods of artificial recharge of groundwater.	L2	CO5	10
13.	a.	How the different rocks are interrelated?	L4	CO2	10
		OR			
	b.	Explain the classification of Folds with neat sketch. What are the effects of folding in civil engineering projects?	L2	CO4	10